

C)	If $f(x) = \begin{cases} x & , \quad 0 < x < \frac{\pi}{2} \\ \pi - x & , \quad \frac{\pi}{2} < x < \pi \end{cases}$ then find half range Fourier sine series Hence show that $f(x) = \frac{4}{\pi} \left(\sin x + \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} + \dots \right)$	Understand (CO4)	6
Q. 5	Solve Any Two of the following.		12
A)	If $\vec{r} = xi + yj + zk$ and $r = \vec{r}$ then Find $\nabla \cdot \vec{F}$, where $\vec{F} = \left(\frac{x}{r}\right)i + \left(\frac{y}{r}\right)j + \left(\frac{z}{r}\right)k$	Understand (CO5)	6
B)	Verify Green's theorem for $\oint_C (xy + y^2)dx + x^2dy$ where C is bounded by $y = x$ and $y = x^2$	Understand (CO5)	6
C)	Verify the Stokes theorem for $\vec{F} = x^2i + xyj$ over the square in the plane $z = 0$ bounded by the lines $x = 0, x = a, y = 0$ and $y = a$	Apply (CO5)	6
	*** End ***		